

Which Way Did It Move? Diagnosing and Overcoming Directional Motion Blindness in Video-LLMs

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Abstract

Video Large Language Models (Video-LLMs) have made rapid progress on temporal video understanding, yet many fail at a basic perceptual primitive: signed image-plane motion direction. On simple videos of a single object moving left, right, up, or down, most Video-LLMs perform near chance, with above-chance cases largely attributable to prediction biases rather than genuine direction understanding. We call this failure *directional motion blindness*. We localize the failure by tracing motion direction information through the Video-LLM pipeline. Motion direction remains linearly accessible from the vision encoder, projector, and LLM hidden states, but the readout fails to bind this signal to the correct verbal answer option, revealing a *direction binding gap*. Although synthetic motion direction instruction tuning reduces this gap on the source domain, motion direction concept vector analysis shows that visual complexity weakens the signal magnitude and limits out-of-domain generalization. We introduce MODIRECT, a dataset family for motion direction instruction tuning and evaluation, and DeltaDirect, a diagnosis-driven, projector-level objective that predicts normalized 2-D motion vectors from adjacent-frame feature deltas. On MODIRECT-SYNBENCH, instruction tuning with DeltaDirect improves motion direction accuracy from 25.9% to 85.4%. On MODIRECT-REALBENCH, DeltaDirect improves real-world motion direction accuracy by 21.9 points over the vanilla baseline without real-world tuning data, while preserving standard video-understanding performance. The code is available at <https://github.com/KHU-VLL/DeltaDirect>.

1 Introduction

Recent work on Video-LLMs has made rapid progress in temporal understanding [21, 33, 41, 52, 57] with different benchmarks analyzing a model’s temporal capability by testing its memory [51], ability to identify complex human action or object movements [41, 55, 57], identifying the order of stitched videos [41, 80], or understanding long-form videos [16, 44, 82]. Despite this progress, we find that many Video-LLMs [2, 9, 30, 32, 36, 37, 63, 77] fail at a much simpler temporal primitive: motion direction understanding. Consider a seemingly trivial synthetic video in which a yellow circle moves from left to right, as shown in Figure 1. Can a Video-LLM correctly tell which way it moves: left→right, right→left, bottom→top, or top→bottom? To our surprise, most Video-LLMs show accuracy close to random chance (25%), with Gemini2.5-Flash [11] only reaching 53.5% accuracy. Humans identify such motion directions instantly and effortlessly, reflecting the fundamental role of motion direction in visual perception and physical navigation [5, 18, 46]. We refer to this systematic failure to distinguish signed motion directions as *directional motion blindness*.

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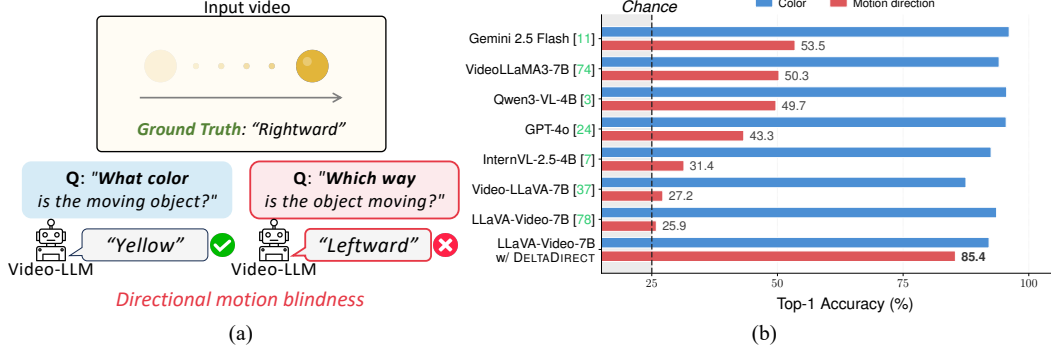


Figure 1: **Directional motion blindness in Video-LLMs.** (a) Given a simple synthetic video of a yellow circle moving from left to right, recent Video-LLMs correctly identify the object’s color but answer the wrong motion direction. (b) Across Video-LLMs, appearance recognition is high, yet signed motion direction accuracy remains much lower, often near chance.

To understand where this failure arises, we trace motion direction information across the Video-LLM pipeline. We find that the signal remains linearly accessible throughout the pipeline: motion direction is decodable from the visual encoder, the projector output, and intermediate LLM states. The breakdown emerges at the readout position, i.e., the final token representation used to predict the answer: although a motion direction signal is still linearly accessible, the model does not reliably connect it to the correct verbal response [45, 49]. We call this mismatch a *direction binding gap*, and observe the same pattern across multiple Video-LLMs (Figure 3). This suggests that directional motion blindness is not simply a failure of visual perception, but a failure to make an available motion signal usable for language-level readout.

We therefore introduce MODIRECT, a dataset family for signed image-plane motion direction, with three subsets: MODIRECT-INST, MODIRECT-SYNBENCH, and MODIRECT-REALBENCH. Its synthetic part contains four domains under a controlled 2×2 design over foreground type and background type: Primitive-on-Syn, Cutout-on-Syn, Primitive-on-Real, and Cutout-on-Real. Primitive videos contain rendered geometric primitives such as circles, triangles, and squares; Cutout videos contain segmented real-world object images [38]. Synthetic videos use uniform-color backgrounds, whereas Real videos use natural scene backgrounds [81]. From the simplest synthetic domain, Primitive-on-Syn, we construct MODIRECT-INST for instruction tuning. We use all four synthetic domains as MODIRECT-SYNBENCH for controlled synthetic evaluation, and use MODIRECT-REALBENCH for real-video evaluation.

Instruction tuning on MODIRECT-INST substantially improves on the source domain motion direction recognition. However, models that perform well on Primitive-on-Syn still degrade on more complex domains, especially Cutout-on-Real. This indicates that next-token prediction on motion direction instructions can improve answer binding on the source domain, Primitive-on-Syn, but does not necessarily produce a strong domain-invariant motion direction signal.

We next ask why this generalization gap persists. Using difference-in-means [1, 31, 45, 53, 60] motion direction concept vectors, we extract motion direction concept vectors from the readout position of each LLM layer for each domain and motion direction. After instruction tuning, motion direction concept vectors become well aligned across domains, despite being trained only on Primitive-on-Syn; for example, at certain layers, their cross-domain cosine similarity is much higher than in the baseline without motion direction instruction tuning. Yet their magnitude drops sharply on complex domains, mirroring the observed accuracy drop. Moreover, restoring only the magnitude of the motion direction concept vector recovers much of the lost accuracy. These results suggest that the out-of-domain failure is not due to a missing motion direction geometry, but to a *magnitude deficit*: the model learns the motion direction structure, but the signal is too weak to be reliably read out across domains. Please refer to Section 3 for more details.

This diagnosis motivates DeltaDirect, a training-only projector-level objective that strengthens signed displacement cues at the visual-language interface. Rather than adding motion tokens or a motion-specific encoder at inference time, DeltaDirect uses analytically available 2-D motion vectors from MODIRECT-INST as supervision during training. Specifically, DeltaDirect predicts normalized 2-D motion vectors from adjacent-frame projector-feature deltas, encouraging the

projector output to carry a stronger signed displacement signal before it enters the LLM. The auxiliary branch is discarded after training, leaving the test-time input format, token sequence, model architecture, and decoding procedure unchanged.

We evaluate `DeltaDirect` across synthetic, real-world, and general video understanding benchmarks. On `MODIRECT-SYBENCH`, instruction tuning on `MODIRECT-INST` improves over the zero-shot model, and adding `DeltaDirect` further improves average motion direction accuracy by 6.5 points, with the largest gain of 11.2 points on `Cutout-on-Real`. To test real-world transfer, we construct `MODIRECT-REALBENCH`, a motion direction benchmark curated from `Something-Anything-V2 (SSv2)` [19], `TOMATO` [57], and `KTH` [56]. Without seeing real-world videos during tuning, `DeltaDirect` improves accuracy by 21.9 points over the vanilla baseline. Beyond motion direction-specific benchmarks, `DeltaDirect` preserves or improves performance on standard video understanding benchmarks [21, 32, 41, 44, 51, 61, 75], suggesting positive transfer rather than overfitting to synthetic motion. Finally, when applied during full fine-tuning, `DeltaDirect` alone yields strong improvements on both motion direction and general video benchmarks, indicating that motion-vector supervision can serve as a useful training signal for future Video-LLMs.

In this work, our major contributions are threefold:

- We identify *directional motion blindness* in Video-LLMs: despite strong appearance recognition, recent models fail to distinguish basic signed motion directions such as left, right, up, and down. We diagnose this failure as a *direction binding gap*, where motion direction information is linearly decodable but not reliably bound to the correct verbal response.
- We show that instruction tuning closes the direction binding gap on the source domain but leaves an out-of-domain generalization gap. Through concept vector analysis, we trace this gap to a magnitude deficit in shared motion direction representations across domains.
- We introduce `MODIRECT`, a controlled dataset family for motion direction instruction tuning and evaluation, and propose `DeltaDirect`, a training-only auxiliary objective that predicts motion vectors from adjacent-frame feature deltas. `DeltaDirect` improves motion direction understanding while preserving or improving general video understanding.

2 Related Work

Video-LLMs and temporal-motion benchmarks. Video-LLMs extend image-based MLLMs to temporal understanding through video instruction tuning, unified image-video architectures, and long-video modeling [2–4, 9–11, 24, 30, 36, 37, 39, 42, 43, 63, 74]. Existing benchmarks evaluate temporal memory, action and event understanding, temporal ordering, and long-form video reasoning [12, 32, 41, 44, 57]. Motion direction appears in several temporal or motion-centric benchmarks, often alongside speed, order, rotation, and broader motion reasoning [14, 21, 34, 61, 67, 75, 83]. These benchmarks are valuable for real-world evaluation, but real videos entangle apparent direction with camera motion, occlusion, object identity, scene layout, scale change, and event semantics. We instead separate diagnosis from transfer: controlled synthetic videos *isolate signed image-plane direction*, and real-world benchmarks test whether the intervention transfers.

Mechanistic analysis of LLMs. A growing line of work analyzes internal representations rather than final outputs alone. Probing and difference-in-means analyses show that model states can encode information that is not reliably verbalized [45, 49]. For Video-LLMs, recent work begins to trace layer-wise information flow and temporal bottlenecks inside the model [27, 33]. Our analysis is complementary: we isolate motion direction as a single perceptual primitive and ask *whether the signal is present, where it weakens, and how it couples to verbal readout*.

Motion representations and supervision. Video models encode motion explicitly through frame differences, optical flow, or trajectories for action recognition [29, 35, 62], flow distillation [13, 58, 84], and self-supervised learning [23, 69]. Recent Video-LLMs use motion-vector tokenization [26], compressed-video cues [79], optical flow [40, 47], or motion-appearance disentanglement [73]. `DeltaDirect` instead uses motion vectors only as training-time supervision on projector-feature deltas, without changing the test-time input or decoding pipeline.

3 Where Does Directional Motion Blindness Arise in Video-LLMs?

We investigate a Video-LLM, `LLaVA-Video` [78], from visual representations to the answer token, and test whether the failure is due to missing supervision, insufficient prompting, visual encoding, projection, or the final binding between a perceived direction and the corresponding answer option. We also investigate several Video-LLMs and draw the same conclusions. For more results and details, refer to Appendix D.

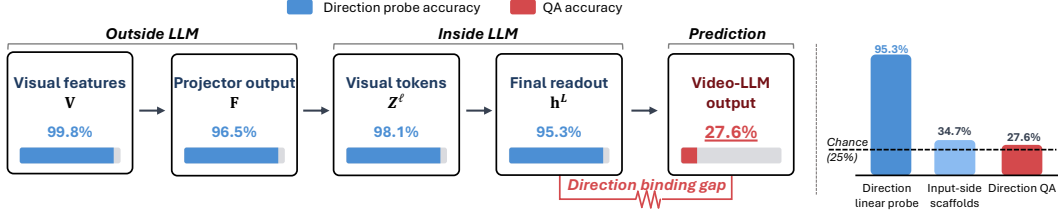


Figure 2: **Direction is decodable, but not converted into the answer.** On Primitive-on-Syn in MODIRECT-SYNBENCH, motion direction remains linearly decodable throughout LLaVA-Video-7B [78], yet QA accuracy stays near chance, exposing the *direction binding gap*.

We organize the analysis around three research questions. (i) Can the failure be explained by missing direction supervision or insufficient input-side scaffolding? (Section 3.2) (ii) Is the signed direction erased by the vision encoder or the vision-language projector, or the intermediate LLM layers? (Section 3.3) (iii) If direction remains accessible, can the final readout token bind it to the prompt-specific answer option? (Section 3.4) We find that motion direction remains linearly accessible across the visual encoder, projector, and LLM hidden states, but the final readout token fails to encode the mapping from direction to answer option. We call this mismatch a *direction binding gap*.

3.1 Experimental Setup

Datasets. We introduce MODIRECT-SYNBENCH, a controlled synthetic benchmark for signed image-plane motion direction. MODIRECT-SYNBENCH follows a 2×2 design over foreground type and background type: Primitive-on-Syn, Cutout-on-Syn, Primitive-on-Real, and Cutout-on-Real. Each video contains a single foreground object moving in one of four signed directions: left, right, up, down. We randomize start and end positions within the valid frame region and use held-out start-end trajectories for evaluation, reducing static-position shortcuts. In every analysis on MODIRECT-SYNBENCH, we train a linear probe on the training split and evaluate it on held-out videos.

Task. We use a four-way multiple-choice question (MCQ) format as the primary evaluation protocol. The four options correspond to left→right, right→left, bottom→top, and top→bottom. Crucially, we randomize the order of answer options for each example. Therefore, the correct answer letter is not a fixed alias of the motion direction; the model must bind the perceived direction to the prompt-specific option text. This design enables single-token, logit-based evaluation and reduces free-form generation confounds, while directly testing whether the model can bind the perceived direction to the correct prompt-specific option [27, 32, 54].

Video-LLM pipeline. A Video-LLM processes a video through three stages. Given T sampled frames, a vision encoder produces visual features $\mathbf{V} \in \mathbb{R}^{T \times M \times D_v}$, where M denotes the number of patch tokens per frame and D_v denotes the embedding dimension. A projector maps these features into the LLM embedding space, producing visual tokens $\mathbf{F} \in \mathbb{R}^{T \times N \times D}$, where $N \leq M$ reflects spatial pooling and D is the LLM hidden dimension. An L -layer LLM consumes the visual tokens with text tokens and produces an answer. Inside the LLM, we track the hidden states at the visual-token positions across layers. Let $\mathbf{z}_t^\ell \in \mathbb{R}^D$ denote the spatially pooled hidden state of the visual-token positions at frame t and layer $\ell \in \{1, \dots, L\}$.

We refer to the hidden state at the last token position used for next-token prediction as the *readout token*, and denote $\mathbf{h}^\ell \in \mathbb{R}^D$ for the readout token at layer ℓ . We call $\mathbf{h}^L$ the *final readout token*; this is the representation the Language Model (LM) head maps to output vocabulary logits.

3.2 Can Data or Prompting Explain the Failure?

Before inspecting model internals, we rule out two external explanations. First, motion-direction supervision is scarce in existing video instruction data. Using keyword pre-filtering followed by semantic classification, we estimate that only 0.91% of LLaVA-Video-178K [78] examples are direction-dependent; human verification further indicates that these estimates are upper bounds (Appendix D.1). This scarcity motivates controlled motion-direction supervision, but it does not localize where the failure occurs. Second, simple input-side scaffolds do not solve the task. We test visual boundary cues, step-by-step location reasoning [28, 64], and coordinate-grid prompts [6, 8, 71]; the best combination reaches only 34.7% on Primitive-on-Syn as shown in Figure 2. Thus, external factors alone are insufficient, and we turn to representation-level diagnostics.

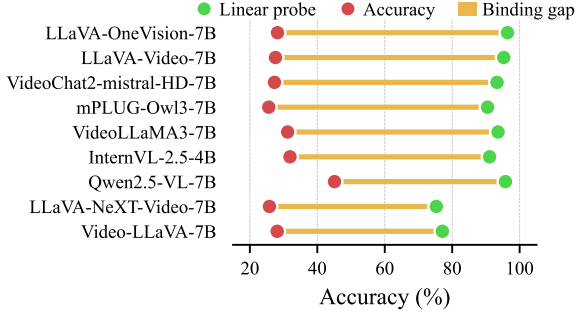


Figure 3: The direction binding gap is universal across Video-LLMs.

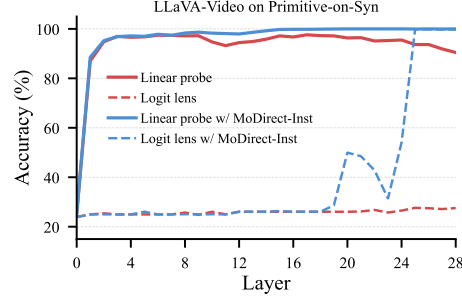


Figure 4: Instruction tuning closes the direction binding gap (source domain).

3.3 Is Motion Direction Erased by the Vision Encoder or Projector?

We next ask whether the failure is caused by visual processing. We use linear probing as a diagnostic tool [20, 31, 49]: if a linear classifier can recover the signed direction from a frozen representation, then that representation contains direction information in a linearly accessible form. This does not imply that the model can directly use the signal for generation; rather, it tests whether the visual-temporal evidence is present before the final readout.

Vision encoder preserves direction. We train a four-way linear probe on the frozen vision-encoder output $\mathbf{V}$. The probe recovers motion direction with 99.8% accuracy (Figure 2), indicating that the vision encoder preserves a near-perfect direction signal.

Projector also preserves direction. We repeat the same probe on the projector output $\mathbf{F}$. Accuracy remains high at 96.5%, indicating that the vision-language projection step does not erase the signed direction signal. Thus, directional motion blindness is not explained by a lack of linearly accessible direction information in either the vision encoder or the projector. The failure must arise from *how this signal is used or read out by the LLM*.

3.4 Can the Readout Bind Motion Direction to the Answer Option?

Direction information reaches the LLM input in a linearly accessible form. We now ask whether it remains accessible inside the LLM and whether the readout token binds this direction signal to the correct prompt-specific answer option.

Direction remains decodable inside the LLM. We probe two internal representations across LLM layers: the visual tokens $\mathbf{z}_t^\ell$ and the readout token $\mathbf{h}^\ell$. As shown in Figure 2, visual-token probes achieve $98.1\% \pm 1.3$, and readout token probes achieve $95.3\% \pm 2.4$. These results show that the LLM does not simply discard the direction signal.

The bottleneck is answer-option binding. The key question is whether the internally decodable direction signal reaches the correct *answer option*. Because the MCQ option order is randomized, the answer letter is not a fixed alias of the motion direction; the model must bind the perceived direction to the option text in the current prompt. Yet the model achieves only 27.6% MCQ accuracy, near the 25% chance level (Figure 2). This shows that the model carries the direction signal internally but fails to bind it to the prompt-specific answer option.

Direction binding gap. Taken together, these results show that the bottleneck lies not in representing motion direction, but in converting that representation into the correct answer option. The direction signal remains accessible across the vision encoder, the projector, and the readout state, yet it is not reliably bound to the prompt-specific answer option [49, 50, 59]. We call this mismatch the *direction binding gap*. As shown in Figure 3, this gap persists across architectures and scales, indicating a shared structural limitation rather than a model-specific artifact. We further verify in Appendix D.4 that the same failure appears in open-ended direction-answer generation, confirming that the phenomenon is not an artifact of the MCQ format. This diagnosis motivates Section 4, where we test whether controlled motion direction instruction tuning can close the gap.

4 What Instruction Tuning Solves and What It Does Not

Section 3 localizes directional motion blindness to answer-option binding: direction is linearly accessible inside the model, but the readout does not reliably link it to the correct answer option. We now test the most direct remedy: *motion direction instruction tuning*. We fine-tune on the simplest synthetic domain and ask whether the learned binding transfers to more complex visual conditions.

4.1 Setup: Training on MODIRECT-INST, Testing on MODIRECT-SYNBENCH

Training data. We fine-tune a Video-LLM on MODIRECT-INST, a synthetic instruction dataset built from the Primitive-on-Syn domain. MODIRECT-INST contains 100K training videos with diverse motion direction-related QA formats, including direction MCQ, open-ended direction, and static appearance questions (e.g., object color and shape, to preserve general visual recognition).

Evaluation data. We use MODIRECT-SYNBENCH as our testbed. MODIRECT-SYNBENCH includes Primitive-on-Syn, the source domain used to construct MODIRECT-INST, and three OOD domains: Primitive-on-Real, Cutout-on-Syn, and Cutout-on-Real. These OOD domains introduce natural scene backgrounds [81], realistic object cutouts [38], or both. This split allows us to distinguish source domain binding from domain-invariant direction understanding.

Training. We use LLaVA-Video-7B [78] as the backbone, keep the vision encoder and original LLM weights frozen, and update only the projector and LoRA [22] adapters in the LLM. We provide the comprehensive details of the analysis in Appendix A.3.

4.2 Source Domain: Answer-Option Binding Emerges

We evaluate whether instruction tuning closes the direction binding gap on the held-out Primitive-on-Syn test set. We repeat the layer-wise probing analysis from Section 3. To diagnose answer-option binding, we use the *logit lens* [27, 48], which projects each intermediate readout token $\mathbf{h}^\ell$ through the LM head without training an external classifier. This measures whether the LM head can recover the correct answer letter at each layer. As shown in Figure 4, in the vanilla model, direction is highly decodable from the readout token (—), but its logit lens accuracy stays near chance (- -), matching its low MCQ accuracy of 27.6%. After instruction tuning, logit lens accuracy (- -) rises in the late layers and converges to direction probing accuracy (—) in the final layers. MCQ accuracy correspondingly increases to 99.5%. Thus, direction-focused instruction tuning establishes a source domain binding pathway from the internal direction signal to the correct answer option.

4.3 Out-of-Domain: The Binding Gap Reopens

We next ask whether this learned binding transfers to more complex visual conditions. As shown in Figure 5 (a), instruction tuning nearly closes the gap on the source domain Primitive-on-Syn test set (0.3 points), but the gap reopens on OOD domains, e.g., 12.1 points on Cutout-on-Real. Although direction remains decodable from the final readout token, MCQ accuracy drops as object and background complexity increase. This suggests that next-token supervision learns source domain answer binding, but does not produce a direction signal strong enough for reliable OOD readout.

4.4 Why Does the Binding Gap Reopen? Shared Orientation but Insufficient Magnitude

To understand the OOD failure, we analyze *motion direction concept vectors* using difference-in-means [1, 31, 45, 53, 60] at the readout state after instruction tuning. For each domain A , direction class d , and layer ℓ , we define $\mathbf{v}_{d,A}^\ell = \mathbb{E}[\mathbf{h}^\ell \mid y = d, A] - \mathbb{E}[\mathbf{h}^\ell \mid A]$. Here we compute the expectation over the MODIRECT-SYNBENCH samples. We decompose each vector into its unit orientation $\hat{\mathbf{v}}_{d,A}^\ell = \mathbf{v}_{d,A}^\ell / \|\mathbf{v}_{d,A}^\ell\|$ and magnitude $\|\mathbf{v}_{d,A}^\ell\|$. We use “orientation” here to distinguish the representation-space axis from the physical motion direction.

Orientation aligns across domains. To test whether the representation space orientation of each motion direction is shared across domains, we compare unit motion direction concept vectors across domains. For each layer ℓ , direction $d \in \{\text{left}, \text{right}, \text{up}, \text{down}\}$, and unordered pair of domains (A, B) , we compute $\cos(\hat{\mathbf{v}}_{d,A}^\ell, \hat{\mathbf{v}}_{d,B}^\ell)$. Since MODIRECT-SYNBENCH has four domains, this gives $\binom{4}{2} = 6$ domain pairs and therefore $4 \times 6 = 24$ cosine similarity values per layer. The cosine similarity reported in Figure 5 (b) is the average over these values. After instruction tuning, this average rises sharply in the late layers, indicating that the model learns a shared motion direction concept vector orientation for each motion direction across domains, despite being trained on Primitive-on-Syn.

Magnitude drops under visual complexity. The aligned orientation alone is insufficient. As shown in Figure 5 (c), the same late layers exhibit a magnitude drop under increasing visual complexity; in particular, Cutout-on-Real shows a clear magnitude deficit relative to Primitive-on-Syn. This mirrors the OOD MCQ accuracy drop and suggests a *magnitude deficit*: although domains

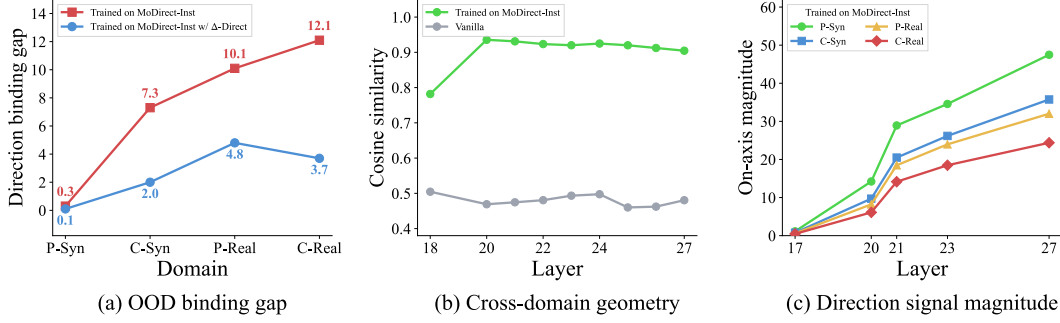


Figure 5: **Shared orientation, weak magnitude.** (a) Instruction tuning closes the binding gap on Primitive-on-Syn, but the gap reopens on OOD domains; DeltaDirect narrows it across domains. (b) Direction concept vector orientations align across domains after instruction tuning, with late-layer cosine similarity exceeding 0.9. (c) Despite this alignment, concept-vector magnitude decreases with visual complexity, revealing the *magnitude deficit* behind the OOD binding gap.

share a motion direction concept vector orientation for each motion direction, the corresponding signal magnitude is too weak in complex domains to support reliable answer-option binding.

Restoring magnitude recovers accuracy. As a diagnostic intervention, for each OOD sample with ground-truth motion direction d , we rescale $\mathbf{h}^\ell$ along the corresponding unit concept-vector orientation $\hat{\mathbf{v}}_{d,A}^\ell$ to match the average Primitive-on-Syn magnitude, leaving the orthogonal component unchanged. We observe that this intervention improves OOD MCQ accuracy, with the largest gain of 15.5 points on Cutout-on-Real. These results suggest that the OOD binding gap is not caused by a missing shared orientation across domains, but by *insufficient signal magnitude*. For more results and details, refer to Appendix D.8.

This motivates a training objective (Section 5) that does not merely teach the answer token, but makes the projector output carry a stronger signed displacement signal before it enters the LLM.

5 DeltaDirect: Strengthening Motion Signals at the VL Interface

The preceding sections suggest a specific intervention point. Directional motion blindness is not caused by the complete absence of direction evidence: direction is linearly accessible from the vision encoder, projector output, and LLM hidden states (Section 3). However, this signal is not reliably bound to the correct answer option. Moreover, motion direction instruction tuning closes the binding gap on the source domain, but visual complexity weakens the magnitude of the shared concept-vector orientation, reopening the gap in out-of-domains (Section 4).

These observations motivate a simple design principle: make *the projector output carry a stronger signed displacement signal* before it enters the LLM. We therefore introduce DeltaDirect, a training-only auxiliary objective applied to the projector output. Instead of adding learned motion tokens, or a motion-specific encoder at inference time, DeltaDirect uses synthetic 2-D motion vectors as supervision during training. The auxiliary branch is discarded after training, so the test-time input format, token sequence, model architecture, and decoding procedure remain unchanged.

These observations lead to three design requirements. First, the supervision should act at the projector output, where visual tokens enter the LLM. Second, it should target motion-induced change rather than static appearance, so we supervise adjacent-frame feature deltas. Third, it should supervise signed displacement directly, rather than only the answer word, so we predict normalized 2-D motion vectors. DeltaDirect instantiates these requirements with a training-only branch.

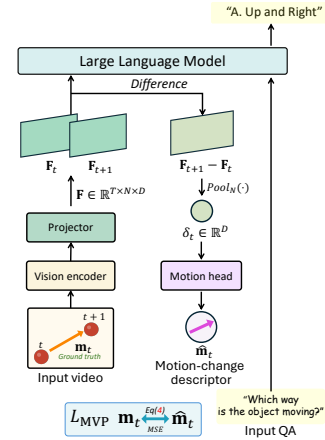


Figure 6: **DeltaDirect.**

5.1 Motion-Induced Feature Deltas

`DeltaDirect` operates on the projector output $\mathbf{F} \in \mathbb{R}^{T \times N \times D}$. For each adjacent frame pair, we spatially pool the temporal feature difference to obtain a motion-change descriptor:

$$\delta_t = \frac{1}{N} \sum_{n=1}^N (\mathbf{F}_{t+1}[n] - \mathbf{F}_t[n]) \in \mathbb{R}^D, \quad t = 1, \dots, T-1. \quad (1)$$

Feature differencing suppresses content that remains stable across adjacent frames and focuses the auxiliary branch on motion-induced change. We compute deltas at the projector output because these are the visual tokens directly passed to the LLM. This lets the auxiliary branch emphasize motion-induced changes in the same representation space used for language readout, without modifying the vision encoder or adding inference-time motion inputs.

5.2 Training-Only Motion Vector Prediction

Motion vector target. Because `MODIRECT-INST` is synthetic, the object center $\mathbf{c}_t \in \mathbb{R}^2$ is known at each frame. For each moving frame pair, we define the normalized 2-D motion target

$$\mathbf{m}_t = \frac{\mathbf{c}_{t+1} - \mathbf{c}_t}{\|\mathbf{c}_{t+1} - \mathbf{c}_t\|_2 + \varepsilon} \in \mathbb{R}^2. \quad (2)$$

This target captures the signed image-plane displacement direction while removing speed as a confounding factor. Rather than treating directions as discrete labels for an auxiliary classifier, the normalized vector target preserves the geometry of signed displacement: opposite motions have opposite targets, and different directions remain geometrically related in the same 2-D space.

Prediction head. A lightweight linear head predicts the motion vector from each pooled delta:

$$\hat{\mathbf{m}}_t = \mathbf{W} \delta_t + \mathbf{b}, \quad \mathbf{W} \in \mathbb{R}^{2 \times D}, \quad \mathbf{b} \in \mathbb{R}^2. \quad (3)$$

The head introduces only $2D + 2$ trainable parameters and is used only during training.

MVP loss. We define the Motion Vector Prediction loss as

$$L_{\text{MVP}} = \frac{1}{T-1} \sum_{t=1}^{T-1} \|\hat{\mathbf{m}}_t - \mathbf{m}_t\|_2^2. \quad (4)$$

This objective does not ask the model to memorize a direction word. Instead, it directly shapes the projector output so that adjacent-frame changes preserve signed displacement cues before entering the LLM. We apply MVP supervision at the projector output by default, and compare it with intermediate-layer and readout-position supervision in Appendix E.

5.3 Training Objective and Inference

The final training objective combines standard next-token prediction with the MVP loss:

$$L_{\text{total}} = L_{\text{LM}} + \lambda L_{\text{MVP}}, \quad (5)$$

where L_{LM} is the next-token cross-entropy on answer tokens and λ controls the auxiliary loss weight. The MVP branch updates only the projector and prediction head; the LLM decoder is trained solely by L_{LM} . During inference, the MVP branch is removed, leaving the original input format, token sequence, and decoding procedure unchanged.

6 Experimental Results

We evaluate whether `DeltaDirect` turns our diagnosis into practical gains. We focus on three questions: (i) Does it reduce the diagnosed binding gap? (ii) Does `DeltaDirect` improve synthetic and real-world motion-direction understanding without real-world direction supervision? (iii) Does it preserve general video understanding? Implementation details, benchmark protocols, and additional analyses and ablations are provided in Appendix A, E.

6.1 Improving Motion Direction without Sacrificing General Video Understanding

The direction binding gap narrows. In Figure 5 (a), instruction tuning on `MODIRECT-INST` leaves a widening binding gap (12.1 on Cutout-on-Real) under visual complexity. `DeltaDirect` reduces this gap to 3.7 points on Cutout-on-Real and narrows it on every OOD domain.

Table 1: **DeltaDirect** achieves state-of-the-art motion direction understanding. We report Top-1 accuracy (%) on MODIRECT-SYNBENCH and MODIRECT-REALBENCH.

Method	MODIRECT-SYNBENCH					MODIRECT-REALBENCH				Overall Avg.
	P-Syn	C-Syn	P-Real	C-Real	Avg.	SSv2 [19]	KTH [56]	TOMATO [57]	Avg.	
Random Chance	25.0	25.0	25.0	25.0	25.0	50.0	50.0	20.0	40.0	31.4
GPT-4o [24]	46.6	37.8	48.0	40.7	43.3	51.0	52.3	34.5	45.9	44.4
Gemini 2.5 Flash [11]	58.1	43.4	61.7	50.6	53.5	20.0	69.4	25.6	38.3	47.0
Video-LLaVA-7B [37]	28.1	27.8	25.5	27.2	27.2	49.2	49.5	14.6	37.8	31.7
VideoChat2-HD-7B [32]	26.1	23.5	24.6	24.1	24.6	50.0	54.2	20.1	41.4	31.8
LLaMA-VID-7B [36]	25.3	25.6	24.7	25.0	25.2	51.4	54.2	16.9	40.8	31.9
LLaVA-NeXT-Video-7B	25.8	24.8	25.0	25.2	25.2	50.8	52.4	21.1	41.4	32.2
LLaVA-OneVision-7B [30]	28.3	23.4	29.9	29.3	27.7	52.2	50.3	26.8	43.1	34.3
Qwen2.5-VL-7B [4]	45.1	32.5	30.8	30.2	34.7	52.6	44.2	25.3	40.7	37.2
Qwen3-VL-4B [3]	66.6	50.2	40.8	41.1	49.7	60.9	62.1	35.0	52.7	51.0
InternVL-2.5-4B [7]	31.9	31.8	30.4	31.4	31.4	66.1	50.7	26.1	47.6	38.3
VideoLLaMA3-7B [74]	56.4	46.6	50.0	48.3	50.3	56.5	52.2	19.9	42.9	47.1
LLaVA-Video-7B [78]	27.6	23.4	26.9	25.8	25.9	52.2	50.3	26.8	43.1	33.3
LLaVA-Video-7B w/ FlashVID [15]	25.2	25.1	24.3	24.8	24.9	52.5	50.3	21.8	41.5	32.0
LLaVA-Video-7B w/ MODIRECT-INST	99.5	80.7	74.7	60.5	78.9	72.4	66.6	35.2	58.1	69.9
LLaVA-Video-7B w/ DeltaDirect	99.7	84.9	85.2	71.7	85.4	81.5	74.8	38.8	65.0	76.7
Qwen2-0.5B Full-FT w/ MODIRECT-INST	99.5	97.3	62.5	51.5	77.7	59.0	55.2	20.6	44.9	63.7
Qwen2-0.5B Full-FT w/ DeltaDirect	99.7	98.7	91.0	80.1	92.4	77.8	69.4	21.7	56.3	76.9

Table 2: Where to apply supervision.

Stage	MODIRECT		MVBench
	SYNBENCH	REALBENCH	
None	78.9	58.1	60.9
Vision encoder	80.4	59.3	60.9
Pre-projector	86.1	62.2	60.2
Post-projector	85.4	65.0	60.7
LLM visual tokens	70.0	58.3	57.4
Final readout	27.7	42.5	59.9

Table 3: What motion signal to supervise.

Supervision target	MODIRECT		MVBench
	SYNBENCH	REALBENCH	
MODIRECT-INST tuning only	78.9	58.1	60.9
Frame order	24.7	40.3	54.5
Concat. feature deltas	80.1	58.0	60.7
Delta equivariance	78.4	58.0	60.6
DeltaDirect	85.4	65.0	60.7

DeltaDirect substantially improves motion direction. Table 1 shows that MODIRECT-INST tuning nearly solves Primitive-on-Syn (99.5%) but transfers poorly to the hardest synthetic OOD domain, Cutout-on-Real, (60.5%). **DeltaDirect** targets this gap, improving Cutout-on-Real to 71.7% and raising MODIRECT-SYNBENCH average accuracy from 78.9% to 85.4% over MODIRECT-INST alone. It also transfers to real videos without real-world tuning, improving MODIRECT-REALBENCH Avg. from 43.1% to 65.0% over vanilla LLaVA-Video-7B. In full fine-tuning, the same trend holds: **DeltaDirect** raises MODIRECT-SYNBENCH performance from 77.7% to 92.4% and MODIRECT-REALBENCH Avg. from 44.9% to 56.3%.

DeltaDirect improves general video understanding.

Although **DeltaDirect** targets motion direction, it improves aggregate general video performance in Standard [25, 32, 44, 51, 66] and Fine-grained [21, 41, 61, 75] video benchmarks. For LLaVA-Video-7B, Standard and Fine-grained averages increase from 69.4 to 70.1 and from 47.3 to 48.7, respectively. In the full fine-tuning setup, **DeltaDirect** further improves over MODIRECT-INST on both averages (58.7 $\rightarrow$ 59.4, 32.8 $\rightarrow$ 33.3). Thus, direction supervision does not trade off broader video capability.

Table 4: General video understanding results.

Method	Standard	Fine-grained
LLaVA-Video-7B [78]	69.4	47.3
LLaVA-Video-7B w/ DeltaDirect	70.1	48.7
Qwen2-0.5B Full-FT w/ MODIRECT-INST	58.7	32.8
Qwen2-0.5B Full-FT w/ DeltaDirect	59.4	33.3

6.2 Ablation Study

Where to apply supervision. In Table 2, we apply MVP supervision at various locations in a Video-LLM. Post-projector supervision performs best on average, while readout supervision performs poorly despite the binding gap being observed there. This supports strengthening motion signals at the vision-language interface, before the LLM transforms them for answer generation.

What motion signal to supervise. In Table 3, we compare **DeltaDirect** with various supervision targets including a direction-aware delta equivariance objective. **DeltaDirect** performs best, indicating that signed 2-D motion-vector supervision is most effective.

7 Conclusions

In this work, we identify *directional motion blindness*, a systematic failure of Video-LLMs to resolve signed image-plane motion direction. Through controlled diagnosis on MODIRECT, we localize the

failure to a *direction binding gap*: motion direction remains linearly accessible inside the model, but is not reliably connected to the correct verbal response. Instruction tuning closes this gap on the source domain, but visual complexity reopens it through a magnitude deficit in the shared direction representation. We address this with `DeltaDirect`, a training-only objective that predicts motion vectors from adjacent-frame projector-feature deltas, strengthening signed displacement cues without changing inference. Trained only with synthetic motion supervision, `DeltaDirect` improves synthetic and real-world direction understanding while preserving general video understanding. These results suggest that diagnosis-driven supervision can help close perception–language gaps in Video-LLMs.

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